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Modelling port competition for intermodal network design with environmental concerns

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ABSTRACT

In the fast-moving and competitive market conditions, port operators need to improve their competitiveness to attract more customers. With the rise of stakeholders' (e.g., shippers and port operators) environmental concerns, intermodal transport offers an opportunity to reduce environmental impacts and improves port competitiveness, due to the benefit on environmental protection. However, it is a challenge to understand how port operators make their plans for intermodal network design with consideration for port competition, shippers' route choice behaviors and environmental concerns of stakeholders. In this paper, we develop a game-theoretical model of port competition with environmental concerns. A logit model is used to formulate shippers' discrete choice behaviors. With respect to the probability of nonexistence of pure Nash equilibrium, we propose an approximate equilibrium and a corresponding solution algorithm. A case study on the competition between Dalian port and Yingkou port in the northeast of China is used to verify the effectiveness of the proposed model and algorithm. Results show that, in competitive settings, the rise of stakeholders' environmental concerns will make the market less competitive and always be beneficial to the port closer to the market (here Yingkou port). Moreover, when shippers pay more attention to the environment, the total CO_2e emissions will decrease. As the number of dry ports increases, the CO_2e emissions will decrease. There are also some interesting findings of effects of environmental concerns on the intermodal network, profits, and CO_2e emissions.

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1. Introduction

Human expansion of greenhouse effect is commonly believed to be the major cause of dramatic climate changes. The greenhouse gases (GHGs), which contribute to the greenhouse effect, include carbon dioxide (CO_2), methane and Nitrous oxide. In 2013, around 23% of all CO_2 was emitted by transport sectors globally. Take the United States as an example, transport sectors contribute more than 30% of CO_2 emissions and about 85% of these emissions are attributable to road transport. A study commissioned by U.S. Energy Information Administration (EIA), reports that transport sectors will surpass energy sectors and become the biggest contributor before 2020 (EIA, 2017). As the world's second largest economy, China, whose CO_2 emissions has ranked the first globally, is confronted with the enormous pressure to reduce emissions. From the perspective of freight transport, besides alternative fuels and the innovations in vehicles, one of the efficient methods of reducing CO_2 is to promote the use of intermodal transport.

As the average carbon intensity of intermodal transport is 46% lower than truckload, according to the estimation conducted by Craig et al. (2013), intermodal transport makes it possible that freights shift from road transport to rail transport. For this purpose, the rise of environmental concerns will encourage port operators to develop intermodal transport in the hinterland. When designing intermodal network, port operators must consider the port competition. Each port must consider the competitors' strategies under competition. Nowadays port competition is growing severe as the global economy has stepped into a new era characterized by the slower economic growth and weak global trade. In order to keep the competitiveness in the overlapping hinterland, more ports shift their focus to hinterland freight intermodal transport in terms of port regionalization (Notteboom and Rodrigue, 2005; Roso and Lumsden, 2010). A considerable number of dry ports, which play a vital role in the intermodal transport network, are planned and constructed by port operators. From the perspective of shippers, the rising awareness of the need for environmental-friendly

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transport modes may directly influence the freight transport modal choice.

It is a challenge to solve the problem how port operators make plans about intermodal network considering port competition, shippers' route choice behavior, and stakeholders' environmental concerns. Therefore, some research questions will be naturally proposed: How to develop a model to design intermodal transport with the environmental concerns of port operators and shippers under the context of competition? How to find the equilibrium strategies of port operators? What is the effect of the stakeholders' environmental concerns on strategies and profits of port operators and CO_2 emissions?

The intermodal network design problem (INDP), as a special case of hub location problem, has attracted more attention in recent years. The interested readers could refer to Farahani et al. (2013) and Alumur and Kara (2008) for a complete review of hub location problems. Limbourg and Jourquin (2009) focused on their research on the intermodal hub locations on the European network based on p-hub median model. Ishfaq and Sox (2010) proposed a model which extends the p-hub median model for interacting hub location-allocation problems to the domain of intermodal logistic. An increasing number of works have paid attention to the topic of intermodal transport network design. Ambrosino and Sciomachen (2016) proposed the formulation of INDP with capacity bounds which are given both for the candidate hub nodes and arcs. Resat and Turkay (2015) formulated a multi-objective optimization model for the INDP. The work aimed to minimize the transportation cost and time and used an augmented ε-constraint method to solve the problem. Lin et al. (2014) developed a modified model and a heuristic method simplified the models proposed by Sörensen et al. (2012). Ishfaq and Sox (2011) established a model on the basis of the model proposed by themselves in 2010, and encompassed the constraint service time. Then in 2012 they Meng and Wang (2011) considered multiple stakeholders and multiple types of containers and incorporated user equilibrium behavior of intermodal operators in route choice. Wang and Meng (2017) extended their own work (Wang and Meng (2011)) and investigated the intermodal network problem with a random route utility of intermodal operators. The readers interested in INDP could refer the literature review in SteadieSeifi et al. (2014). However, aforementioned studies did not involve the environmental concerns of stakeholders or competition between the network planners. Furthermore, it is still a neglected research question required to be investigated that how to design the intermodal network when the planner is the hub operator.

There exists an emerging trend of studies on intermodal network design with environmental concerns recently. Lam and Gu (2016) developed a bi-objective optimization model to simultaneously minimize cost and transit time with constrained carbon emissions. Demir et al. (2016) proposed a stochastic model to decide robust transportation plans with different objectives, including cost, time and greenhouse gas emissions. In addition, the usage effect of intermodal transport and dry ports on the environment has been investigated by various methods. Discrete-event simulation method was used by Lättilä et al. (2013) to estimate the cost and CO_2e emission development in different scenarios in a Finnish context. Zhang et al. (2013) introduced a bi-level programming model for the optimization of the intermodal network with the cost of CO_2e emissions. The upper-level model searches for the optimal terminal and the lower-level model performs multi-commodity flow assignment. Qu et al. (2016) considered the GHG emissions in the objective function to minimize the total cost and linearized the formulation to find the solution. However, the relevant studies are still scarce, and the works only study the intermodal network design with the environmental concern of planners, the consideration of route choice makers (e.g. shippers) on environment has not thus far appeared in the studies on the classical intermodal network design.

Moreover, a few number of literature focused on the study on the transport network design with competition. Marianov et al. (1999) was the first to study the transport network design with competition. This work assumed that the current player (i.e. the follower) optimized the locations of hubs based on the context where the transport network of the competitor (i.e. the leader) is given. The same assumption was also used by Wagner (2008), Eiselt and Marianov (2009) and Lüer-Villagra and Marianov (2013). Another stream of literature on transport network design with competition was to use a Stackelberg model to investigate the decisions of two planners. In these models, the planners sequentially made a decision on the locations of their hubs. Mahmutogullari and Kara (2016) developed a game model in which both planers aimed to maximize the market share of himself/herself and the customers chose one of the planners to provide transport service after the networks was determined. Sasaki et al. (2014) adopted a generic hub arc model in which the planners located arcs with discounted transport costs connecting pairs of hubs. Sasaki (2005) addressed a game model incorporating flow threshold constraints to void unprofitable services. Lin and Lee (2010) adopted Nash game model to find the equilibrium strategies of both planners. However, those studies did not consider the features of intermodal transport and incorporate the environmental concerns of the stakeholders.

To fill the gap of studies on the intermodal transport network design with environmental concerns and port competition, we develop a game-theoretical model of port competition on the problem proposed in this paper. In the model, each port operator aims to maximize the payoff considering profits and environment. The inland route choice behavior of shippers is formulated by a logit discrete choice model. The environmental concerns of port operators and shippers are considered respectively in the port operator's payoff function and shipper' utility functions. It is also a challenge to find equilibrium strategies of port operators. An equilibrium state means that none of the players would unilaterally alternate their respective strategies to increase profits, which is called a Nash equilibrium. With respect to the possibility of inexistence of pure Nash equilibrium due to the non-differential objective function of the game model, an approximate equilibrium and the corresponding algorithm are proposed based on the works of Grauberger (2015) and Wang et al. (2014). Furthermore, a case study of port competition between Dalian port and Yingkou port is used to verify the proposed model and algorithm. We analyze the effect of environmental concerns on intermodal network design and CO_2e emissions. The result of our analysis has important implications for port management and policymakers.

The paper is structured as follows. Section 2 presents the formulation of intermodal network and introduces CO_2e estimation models. In Section 3, we formulate the problem of intermodal transport network design with environmental concerns and port competition as a Nash game model. An approximate equilibrium and the corresponding algorithm are proposed in Section 4. Section 5 is structured to test the algorithm and analyze the effect of environmental concerns on port competition with a case study. Conclusions and future research directions are stated in Section 6.

2. Intermodal network

2.1. Network formulation

In this subsection, we introduce the presentation of the inland intermodal network. The network is denoted as $G = (N,A)$, where N

is set of nodes and A is set of links. Let M denote the set of transport modes, e.g. M = {road, rail}. A link would exist between the node $i \in N$ and node $j \in N$ with the transport mode $m \in M$. Thus, the set of links is denoted by $A \subset \{a_{ij}^m : i,j \in N, m \in M\}$. Let $P \subset N$ be the set of port nodes. Considering the simplicity and representativeness, P is also used to denote the set of port operators based on the assumption that each port operator just operates one port. Denote $H \subset N$ as the set of candidate nodes of dry port locations and denote $H_p \subset H$ as the set of selected nodes of dry port locations for port operator $p \in P$. We use $A^{road} = \{a_{ij}^m : i,j \in N, m = road\}$ to denote the set of road transport links and use $A^{rail} = \{a_{ij}^m : i \in P, j \in H_p, m = rail\}$ to denote the set of rail transport links, which means the road transport link is available between any two nodes and the rail transport link is available only between a node and a port. We have that $A = A^{road} \cup A^{rail}$.

Fig. 1 shows the inland intermodal network between ports and demand nodes. The transport network is utilized in both directions. Ports are either origins or destinations according to the transportation directions, so are demand nodes. For example, from origin port b to destination demand node v, cargos can be directly transported to v by road, or be transported to dry port q by railway, followed by road transport from q to v. In this paper, it is assumed that ports have the same maritime transport services, since we focus on the inland transport freight and port service which mainly affect the inland route choice of a shipper. The inland freight transport is operated between a demand node and a port. Let $W \subset N$ be the set of demand nodes. For $w \in W$, the set of probable transport routes are denoted as R_w. Denote $R_w^{\text{inter}} \subset R_w$ as the set of probable intermodal transport routes and $R_w^{road} \subset R_w$ as the set of probable road transport routes. Let $\delta_w^{a,r}$ be one if route $r \in R_w$ traverses $a \in A$; otherwise, zero. Let $\delta_w^{b,r}$ be one if route traverses hub $r \in R_w$ traverses $b \in H \cup P$; otherwise, zero. We denote the links and nodes traversed by route r respectively as $A_r = \{a \in A | \delta_w^{a,r} = 1\}$, $\forall r \in R_w$, $w \in W$ and $B_r = \{b \in H \cup P | \delta_w^{b,r} = 1\}$, $\forall r \in R_w$, $w \in W$. We assume that each route only contains one port (i.e. $\sum_{n \in P} \delta_w^{p,r} = 1$, $\forall r \in R_w$, $w \in W$). Thus, for demand nodes $w \in W$, the set of intermodal transport routes can be denoted as $R_w^{inter} = \left\{ r \middle| \sum_{a \in A^{rail}} \delta_w^{a,r} = 1, \delta_w^{p,r} \cdot \delta_w^{h,r} = 1, \sum_{a \in A^{road}} \delta_w^{a,r} = 1, \sum_{p \in P} \delta_w^{p,r} = 1, \sum_{h \in H} \delta_w^{h,r} = 1, \delta_w^{a_{hp}^{rail},r} = 1, \delta_w^{a_{wh}^{road},r} = 1, h \in H_p, p \in P \right\}$, while $R_w^{road} = \left\{ r \middle| \delta_w^{a_{wp}^{road},r} \cdot \delta_w^{p,r} = 1, \sum_{a \in A} \delta_w^{a,r} = 1, \sum_{b \in H \cup P} \delta_w^{b,r} = 1, p \in P \right\}$ denotes the set of road transport routes. And there is $R_w = R_w^{road} \cup R_w^{inter}$. Denote by $D_w \in \mathrm{R}_+ = [0, +\infty)$ the transport demand at nodes $w \in W$. For any $a \in A$, the freight flow is denoted by $v_a \in \mathrm{R}_+$ and for each $w \in W$ and $r \in R_w$, the freight flow is denoted by $f_{w,r} \in \mathrm{R}_+$.

2.2. Environmental impact in intermodal network

The emissions of GHGs from freight transportation is probably the most well-known environmental issue which has been studied due to its global effects (Demir et al., 2015). GHGs absorb and emit radiations within the thermal infra-red range in the atmosphere and significantly raise the Earth's temperature. It is recognized by the United Nations Environment Program (UNEP) that GHGs are responsible for most global warming which would globally lead to more severe droughts and/or floods and the possibility of more extreme weather events. Therefore, the calculation of GHG emissions in different transport modes is very important for the environmental protection and the reduction of GHG emissions in freight transportation sector. With respect to the dominant amount of CO_2 in GHG emissions, the impacts of other GHGs are calculated based on carbon dioxide equivalent (CO_2e). In the following subsections, the used calculation methods of the GHG emissions in different parts of transport process are introduced.

In addition, there is a number of literature on the comparison the environmental impact between intermodal transport and road transport. The interested readers could refer Mostert and Limbourg (2016), Craig et al. (2013), Dekker et al. (2012) and Frémont and Franc (2010).

2.2.1. Road transportation

In road transportation, the amount of emissions is dependent on the energy needed for moving vehicles which can be expressed as its fuel consumption. These factors influence the GHG emissions include vehicles, environment, traffic, drivers, and operation (Demir et al., 2014). The fuel consumption models could be classified into two categories, including macroscopic model and microscopic model. Dozens of the models have been summarized in Demir et al. (2014) and Demir et al. (2011). The microscopic

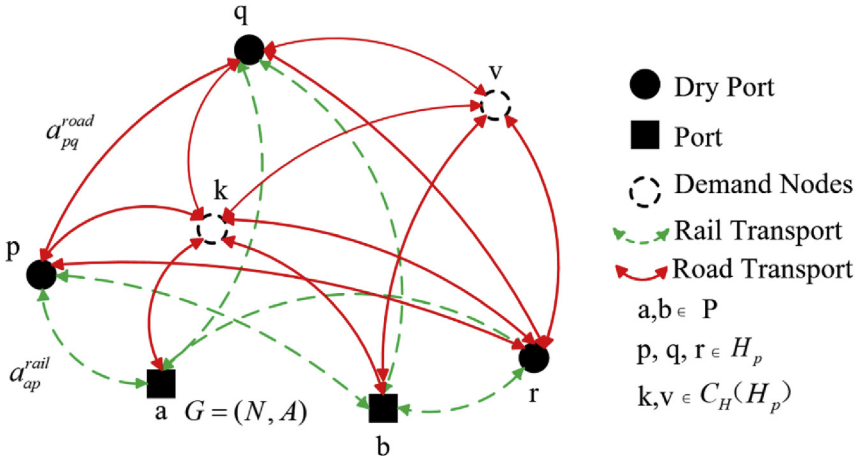


Fig. 1. The inland intermodal network.

models are used to estimate the instantaneous hot-stabilized vehicle emissions and the macroscopic models are usually used in a wide-area emission assessment. Considering the intermodal network design is a strategic planning which does not focus on the instantaneous GHG emissions, we choose COPERT (Computer program to calculate emissions from road transportation) from macroscopic models. The technical development of COPERT is financed by the European Environment Agency (EEA). COPERT uses regression method to estimate fuel consumption which is related to the vehicle class, speed and road type etc. An example of fuel consumption function is given by

$$E_{Ro}(v,D) = (e + (a\exp(-bv)) + (c\exp(-dv)))\cdot D \quad (1)$$

where a to e are the coefficients depending on the type of vehicles, type of fuel, payload and road gradient. Moreover, v denotes vehicle speed (km/h) and D denotes the traveling distance (km).

2.2.2. Rail transportation

The emission models used in rail transportation have also been studied in numerous literature. One of the prominent examples is the calculation framework of EcoTransIt proposed by the Institute for Energy and Environmental Research (IFEU) from Heidelberg. The calculation of energy consumption and emission data based on EcoTransIt can be conveniently done with a web application.[1] The model in this framework is adopted by Demir et al. (2016) and also used as a benchmark model in Kirschstein and Meisel (2015). In this framework, total emissions (i.e. the sum of operation and upstream figures, denoted as Well-to-Wheels/WTW) are calculated based on final emissions (i.e. the emissions caused by operation, denoted as Tank-to-Wheels/TTW) and upstream emissions (i.e. the emissions caused by energy provision, production, and distribution, denoted as Well-to-Tank/WTT). The corresponding formulas applied to calculate final emissions (TTW), upstream emissions (WTT) and total emissions (WTW) are presented as follows.

$$EMV_{tkm} = ECF_{tkm} * EMV_{EC} \quad (2)$$

$$EMU_{tkm} = ECF_{tkm} * EMU_{EC} \quad (3)$$

$$EMT = D\cdot M\cdot (EMV_{tkm} + EMU_{tkm}) \quad (4)$$

where EMT is WTW emissions of transport, and D and M respectively denote the traveling distance and the mass of freight transported. EMV_{tkm} and ECF_{tkm} respectively denote the TTW vehicle emissions and energy consumptions. EMU_{tkm} and ECF_{tkm} respectively denote the WTT vehicle emissions and energy consumptions. EMV_{EC} and EMU_{EC} respectively denotes the emissions coefficient in TTW and WTT.

Based on the statistics, it appears a strong relationship between energy consumption of freight trains and its gross weight in tons, which is presented by

$$EC = 1200 * GWT^{-0.62} \quad (5)$$

where EC is the energy consumption of trains in kilowatt hours and GWT is the gross weight of trains in tons. The function is valid for train weights between 600 and 1800 gross tons which is a typical range for container trains. The function is suitable for hilly countries. For flat countries, the values of the function are multiplied by 0.9, while for mountainous countries the factor is 1.1.

[1] Website: http://www.ecotransit.org/calculation.en.html.

2.2.3. Terminal transfer

Besides the section of road and rail transport, the operation in port and dry port would also contribute the GHG emissions in intermodal transport. Geerlings and van Duin (2011) analyzed the CO_2 emissions from container terminals in the Port of Rotterdam. This work has explicitly investigated the CO_2 emissions of each operation in the port service processes which consume energy. The total energy needed for the movement of equipment is estimated at 18.25 kW h/transshipment of one container from sea ships to railway. This value is also used for seaports in this paper. For the dry ports, the total energy needed is estimated at 4.4 kW h/transfer of one container (IFEU, 2016).

3. Problem design

3.1. Utility of transportation

In this paper, we assume that the shippers are rational, which means the behavior of shippers' route choice depends on their perceived utilities of probable routes. For each route, the utility reflects the shippers' preferences regarding transport cost, transport time and CO_2e emissions. Table 1 shows all the notations and their descriptions in this paper.

The total cost consists of the fare charged by service providers for transporting one unit of cargo (e.g. one TEU) through the links, the nodes and the transport cost. Equation (6) represents the total road transport cost which contains the transportation cost by road and the service charge in port. Equation (7) calculates the total cost for intermodal transport, including the cost of railway transportation from ports to dry ports, the road transportation cost from dry ports to destination, the service charge in ports and dry ports. We use $c_a \in \mathrm{R}^+$ to denote the fare amount for $a \in A$. $\tau^p, \tau_h^p \in \mathrm{R}^+$ are used to denote the service charge for $p \in P$ and $h \in H_p$ respectively. Let C_w^r be the transport cost for any $r \in R_w$, $w \in W$, which is represented by Eqs. (6) and (7).

For road transport:

$$C_w^r = \sum_{a\in A_w^r \cap A^{road}} c_a + \sum_{p\in B_w^r \cap P} \tau_p, \quad \forall r \in R_w^{road}, \ w \in W \quad (6)$$

For intermodal transport:

$$C_w^r = \sum_{a\in A_w^r \cap A^{road}} c_a + \sum_{a\in A_w^r \cap A^{rail}} c_a + \sum_{p\in B_w^r \cap P} \tau_p + \sum_{h\in B_w^r / P, p\in B_w^r \cap P} \tau_h^p, \quad \forall r \in R_w^{inter}, \quad w \in W \quad (7)$$

Equations (8) and (9) represent the total transport time for road transport and intermodal transport respectively. For cargos transported by vehicles only, total time should include the transport time on road and operation time in ports. For intermodal transportation, the transport time includes time on vehicles and trains, and operation time should cover all the operation process in the ports and dry ports. Let T_w^r be the transport time for route $r \in R_w$, $w \in W$.The transport time T_w^r includes the time spent on transferring one unit of cargos (e.g. one TEU) on the links and on operating and waiting at the nodes. Let $t_a \in \mathrm{R}^+$be the transport time for $a \in A$and $t_p, t_h^p \in \mathrm{R}^+$ be the operation time for $p \in P$ and $h \in H_p$. Transport time T_w^r could be calculated as follows.

For road transport:

$$T_w^r = \sum_{a\in A_w^r \cap A^{road}} t_a + \sum_{p\in B_w^r \cap P} t_p, \quad \forall r \in R_w^{road}, \quad w \in W \quad (8)$$

Table 1
Notations and descriptions.

Notation	Description
N	Set of nodes in the transport network
P	Set of ports
H	Set of candidate locations of dry ports
H_p	Set of selected locations of dry port for port operator$p \in P, H_p = \{h \in H \mid y_h^p = 1\}$
W	Set of demand nodes
A	Set of links in the transport network, $A = A^{road} \cup A^{rail}$
A^{road} and A^{rail}	Set of road transport links and set of rail transport links, $A^{road} = \{a_{ij}^m : i,j \in N, m = road\} A^{rail} = \{a_{ij}^m : i \in H_p, j \in P, m = rail\}$
R_w	Set of probable transport routes for node $w \in W$
R_w^{road} and R_w^{inter}	Set of road transport routes and set of intermodal transport routes
A_w^r and B_w^r	Set of links and set of hubs and ports contained in the route $r \in R_w$
M	Set of transport modes $M = \{$road, rail$\}$
D_w	Transport demand at node $w \in W$
a_{ij}^m	The road transport links with m transport modes between node i and node j
y_h^p	Decision variable. If candidate location of dry port $h \in H$ is selected by port$p \in P$; otherwise, zero. $\mathbf{y}^p = \{y_h^p : h \in H\}$
p_1 and p_2	Two competitive ports
$\delta_w^{a,r}$	If route $r \in R_w$ traverses $a \in A$, $\delta_w^{a,r} = 1$; otherwise, zero
$\delta_w^{b,r}$	If route $r \in R_w$ traverses hub $b \in H \cup P$, $\delta_w^{b,r} = 1$; otherwise, zero
τ_p	Service charge of port$p \in P$
τ_h^p	Service charge of dry port $h \in H_p$ operated by port$p \in P$
c_p and C_p	Service cost and fixed cost of port$p \in P$
c_h^p and C_h^p	Service cost and fixed cost of dry port $h \in H_p$ operated by port$p \in P$
c_a	Cost of transporting unit freight by link$a \in A$
C_w^r	Total transport cost per unit freight for route $r \in R_w$, $w \in W$
t_a	Time of transporting freight for link$a \in A$
t^p and t_h^p	Service time of port $p \in P$and service time of dry port located $h \in H_p$ operated by port$p \in P$
T_w^r	Total transport time for route $r \in R_w, w \in W$
e_a	Carbon emission of transporting freight for link$a \in A$
e^p and e_h^p	Carbon emission of port $p \in P$and Carbon emission of dry port located $h \in H_p$ operated by port$p \in P$ for per unit of cargo
E_w^r	Total carbon emission for route $r \in R_w, w \in W$
E_p	Total carbon emission for port$p \in P$
k_p	Number of dedicated berths of port$p \in P$
o_p	Number of ship routes for port $p \in P$
v_p	Service quality of port $p \in P$
z^{road} and z^{inter}	Convenience scores of road transport service and intermodal transport service
σ	Value of convenience
η	Value of dedicated berths
ω	Value of ship routes
γ	Value of service quality parameter
κ and μ	Parameter to capture shippers' and port companies' concerns on environment
ν	Monetary value of carbon emission
ζ	Monetary value of transportation time
ε_w^r	Random terms following Gumbel distribution
U_w^r	Random utility of transporting a unit freight for route $r \in R_w$
V_w^r	Deterministic term of U_w^r
φ_w^r	Probability of choosing route $r \in R_w$for the demands $w \in W$
θ	Scale parameter of the logit model
S_p and S_p^*	Strategy and Nash equilibrium strategy of port $p \in P$
Q_p	Throughput of port $p \in P$
Q_h^p	Throughput of dry port $h \in H$ operated by port $p \in P$
n_p	Quantity of dry ports for port $p \in P$, $n_p = \lvert H_p \rvert$
Π_p	Profit of port $p \in P$
α	Interval factor parameter used in the adapted algorithm
β	Quantity factor parameter used in the adapted algorithm

For intermodal transport:

$$T_w^r = \sum_{a \in A_w^r \cap A^{road}} t_a + \sum_{a \in A_w^r \cap A^{rail}} t_a + \sum_{p \in B_w^r \cap P} t_p + \sum_{h \in B_w^r / P, p \in B_w^r \cap P} t_h^p, \quad \forall r \in R_w^{inter}, \quad w \in W \tag{9}$$

We use E_w^r to denote the amount of CO_2e emissions consumed per unit of cargo (e.g. one TEU) transported on router$\in R_w$, $w \in W$. Equations (9) and (11) define the CO_2e emissions for road transport and intermodal transport respectively. Let $e_a \in R^+$be the amount of CO_2e emissions consumed per unit of cargo (e.g. one TEU) transferred through $a \in A$, which can be calculated based on the methods in section 2.2.1 and 2.2.2 respectively. Let e_p, $e_h^p \in R^+$ be the amount of CO_2e emissions released at port $p \in P$ and dry port $h \in H_p$, which can be calculated based on the reference value selected in section 2.2.3. CO_2e emissions E_w^r could be calculated as follows.

For road transport:

$$E_w^r = \sum_{a \in A_w^r \cap A^{road}} e_a + \sum_{p \in B_w^r \cap P} e_p, \forall r \in R_w^{road}, w \in W \tag{10}$$

For intermodal transport:

$$E_w^r = \sum_{a \in A_w^r \cap A^{road}} e_a + \sum_{a \in A_w^r \cap A^{rail}} e_a + \sum_{p \in B_w^r \cap P} e_p + \sum_{h \in B_w^r / P, p \in B_w^r \cap P} e_h^p, \forall r \in R_w^{inter}, w \in W \tag{11}$$

Besides the considerations presented above, the service quality and service reliability provided by port is also of significance for shippers to choose the transport route and could be presented by the reputation of port. Let k_p and o_p represent the number of berths and shipping routes. v_p represents the service quality, which is scaled from 0 to 10. η, $\omega$and γ are respectively used to transform k_p, o_p and v_p to the value of service quality and service reliability. The degree of convenience of road transport and intermodal transport denoted by $z^{road} \in [0, 10]$ and $z^{inter} \in [0, 10]$. We use σ to transform the degree of convenience to utility of transportation.

From the above, the utility of transportation includes the total cost, transport time, the amount of CO_2e emissions, the degree of convenience and the service quality, and service reliability. For any transport demand node $w \in W$ and for each probable route $r \in R_w$, define U_w^r as the random utility of transportation representing the preference of shipper on route r, and V_w^ras the certainties for U_w^r.

U_w^r can be shown as follows,

For road transport:

$$U_w^r = -C_w^r - \zeta T_w^r - \kappa \nu E_w^r + \sigma z^{road} + \eta k_p + \omega o_p + \gamma v_p + \varepsilon_w^r = V_w^r + \varepsilon_w^r, \forall r \in R_w^{road}, w \in W, p \in B_r \cap P \tag{12}$$

For intermodal transport:

$$U_w^r = -C_w^r - \zeta T_w^r - \kappa \nu E_w^r + \sigma z^{inter} + \eta k_p + \omega o_p + \gamma v_p + \varepsilon_w^r = V_w^r + \varepsilon_w^r, \forall r \in R_w^{inter}, w \in W, p \in B_r \cap P \tag{13}$$

where ν converts the amount of CO_2e emissions to monetary value. κ is a parameter used to reflect shippers' concerns on environmental issues. It is worth noting that if $\kappa > 0$, shippers prefer the route with less amount of CO_2e emissions; if $\kappa = 0$, shippers will

ignore the environmental impact in the transport process. ε_w^r is the stochastic term of U_w^r, which is assumed to capture the unobserved influences and is independent and identically distributed with zero mean value.

3.2. Route choice model

It is necessary to get the probability of choosing the route $r \in R_w$ in $w \in W$. Due to the fact that we only focus on the inland transport network, a choice about inland transport methods is made. Cargos can be transported directly from origin to destination by road, or transported to dry port by train before shipping to destination by road which involves in two modes, called intermodal transport. For all $w \in W$, the probability of shippers choosing the route $r \in R_w$ is defined as

$$\varphi_w^r = \mathrm{P}\left(r \in \underset{j \in R_w}{\operatorname{argmax}}\left\{U_w^j\right\}\right) \tag{14}$$

In this paper, we assume that ε_w^r is an independent and identically distributed random variable following Gumbel distribution. φ_w^r can be represented by a logit model, because of the advantage of charactering the randomness of the utility caused by invisible factors (Wang et al., 2014).

$$\varphi_w^r = \frac{e^{\theta V_w^r}}{\sum\limits_{j \in R_w} e^{\theta V_w^j}}, \forall r \in R_w, w \in W \tag{15}$$

θ is the scale parameter relevant to variance of the stochastic term ε_w^r which complies with Gumbel distribution: $F(\varepsilon_w^r) = e^{-e^{-\theta \varepsilon_w^r}}$. According to the definition of Gumbel distribution, a higher θ value indicates a smaller variance of the stochastic term, which means shippers prefer to choose route with high certain utility.

3.3. Model formulation

In this section, we formulate the port competition model to solve the intermodal network design problem. Each port operator needs to decide the locations of dry ports in order to maximize the payoff of itself. Define the decision variables of port operator $p \in P$ as follows:

$$y_h^p = \begin{cases} 1 & \text{if } h \text{ is selected by port operator } p \text{ as a dry port,} \\ 0 & \text{otherwise,} \end{cases} \quad \forall h \in H, p \in P$$

The set of decision variables about dry port locations is defined as $Y^p = \{y^p \in \{0,1\}^{|H|}\}$ for $p \in P$. The vector $y^p \in Y^p$ denotes a feasible decision of port operator p on the intermodal network's structure. The payoff of port operators simultaneously incorporates the profits and environmental impact which is presented by CO_2e emissions and converted into monetary value. The payoff of port operator $p \in P$is presented as follows:

$$\Pi_p\left(\mathbf{y}^p \middle| \mathbf{y}^{p-}\right) = (1-\mu)\left((\tau_p - c_p)Q_p + \sum_{h \in H}\left(\tau_h^p - c_h^p\right)Q_h^p - C_p - \sum_{h \in H} y_h^p C_h^p\right) - \mu \nu E_p \tag{16}$$

where p is the current decision maker; p^- denotes the competitive port, and it represents different ports in different strategies. Two competitors p_1 and p_2 take turns being current decision maker in game to find an equilibrium strategy. For example, if p_1 is the current decision maker, p represents port p_1 , and p_2 will bep^-. Q_p denotes the throughput of port p, while Q_h^p denotes the throughput of dry port h operated by port p; $E_p$denotes the amount of CO_2e emissions releasing from transporting cargoes during the whole transport process which includes port p. The weight $\mu \in [0,1]$ reflects the port operator's concerns on the environment. The higher value of weight μ implies that port operators put a greater emphasis on the environment when choosing a strategy.

We formulate the problem as a game-theoretical model, a decision problem with multiple decision makers, instead of optimization models which only have a single decision maker. In the model, we consider two competitive players (i.e. port operators). The current player is denoted as p and the competitor is denoted as p^-. Mathematically, the decision behavior of port operator $p \in P$ can be formulated as follows.

$$\max_{\mathbf{y}^p} \Pi_p\left(\mathbf{y}^p \middle| \mathbf{y}^{p-}\right) \tag{17}$$

$$Q_p = \sum_{w \in W}\sum_{r \in R_w} \delta_w^{p,r} \varphi_w^r D_w \tag{18}$$

$$Q_h^p = \sum_{w \in W}\sum_{r \in R_w^{inter}} \delta_w^{h,r} \delta_w^{p,r} \varphi_w^r D_w \, \forall h \in H \tag{19}$$

$$E_p = \sum_{w \in W}\sum_{r \in R_w} \delta_w^{p,r} E_w^r \varphi_w^r D_w \tag{20}$$

$$\sum_{h \in H} y_h^p = n_p \tag{21}$$

$$\varphi_w^r = \frac{e^{\theta V_w^r}}{\sum\limits_{j \in R_w} e^{\theta V_w^j}} \quad \forall r \in R_w, \quad w \in W \tag{22}$$

$$y_h^p \in \{0,1\} \, \forall h \in H \tag{23}$$

where the objective function maximizes the total weighted payoff; constraint (18) and (19) are used to calculate the throughput of the construct dry port and port p; constraint (20) is used to calculate the amount of CO2e emissions of transporting cargoes through port p; constraint (21) ensures that the number of dry ports constructed by port p is fixed; constraint (22) is used to allocate the flows on the probable routes based on a logit model; constraint (23) states the domains of decision variables.

In the competitive environment, each port operator determines a strategy on the intermodal network. The change of competitor's strategy would trigger a reaction and re-purpose a strategy to maximize the weighted payoff. Thus, two players will alternatively choose the strategies, until the game reaches a Nash equilibrium (NE) which means no player has an incentive to change his decision

unilaterally. Let $S_{p_1} = y^{p_1}$ and $S_{p_2} = y^{p_2}$ denote the strategies of port operator p_1 and p_2. $S^*_{p_1}$ and $S^*_{p_2}$ denote the Nash equilibrium with pure strategies of port operator p_1 and p_2. The Nash equilibrium strategy $S^*_{p_1}$ and any other strategies S_{p_1} of the port operator p_1 will satisfy $\Pi_{p_1}(S^*_{p_1}|S^*_{p_2}) \geq \Pi_{p_1}(S_{p_1}|S^*_{p_2})$. The same goes to another port operator p_2, whose Nash equilibrium strategy and any other strategies will also satisfy $\Pi_{p_2}(S^*_{p_2}|S^*_{p_1}) \geq \Pi_{p_2}(S_{p_2}|S^*_{p_1})$.

4. Algorithm

In this section, we discuss the Nash equilibrium solution approach which we could use to solve the ITNDPS model. Considering the probability of non-existence of pure Nash equilibrium solution, we introduce an approximate Nash equilibrium. A corresponding algorithm is proposed to efficiently compute the approximate NE.

Although Nash has proved that according to Brouwer fixed-point theorem, odd equilibrium points with finite mixed discrete strategies can be found for Nash game. However, many games only consider pure strategies rather than mixed ones. For example, in this study, port operators only consider pure strategies, because in real market port operators must make a specific decision. The pure Nash equilibrium solution may not exist in the game with discrete strategy set. The reason is that the players may fall into a close-loop of strategy when choosing the best strategy based on the payoff matrix given the competitor's decision. With respect to the probable non-existence of pure Nash equilibrium solution, Bosse et al. (2007) and Blum and Monsour (2007) proposed a definition of ε-approximate Nash equilibrium. Compared with NE, ε-approximate Nash equilibrium means no player has an incentive to change the current strategy if no other strategy can increase the current payoff by at least $\varepsilon \geq 0$. $S_p^{\varepsilon\,*}$ and $S_{p^-}^{\varepsilon\,*}$ are the ε-approximate NE solutions of port operator p and p- if $\Pi_p(S_p^{\varepsilon\,*}|S_{p^-}^{\varepsilon\,*}) + \varepsilon \geq \Pi_p(S_p|S_{p^-}^{\varepsilon\,*})$ for all strategies. It is obviously found that when $\varepsilon = 0$, the NE is same with ε-approximate NE.

Although the definitions of NE and ε-approximate NE do not appear complicate, it is a hard challenge to find a Nash equilibrium solution even in a simple bi-matrix game, since it has been demonstrated that the problem of computing Nash equilibrium in a bi-matrix game is complete in PPAD (Polynomial Parity Arguments on Directed graphs) as a subclass of NP (Chen et al., 2009). When computing ε-approximate NE, complete payoff matrixes are needed as inputs. However, in the real life, the number of player's feasible strategies may be more than several millions. It would spend considerable time to calculate the players' payoffs with each combination of strategies. Even if the payoff matrixes have been obtained in an available time, it is also a hard work to use an algorithm to find the NE or ε-approximate NE. Therefore, we develop an approximate NE based on the notion of reduced game proposed by Grauberger (2015) and a corresponding algorithm to compute the approximate NE.

One of the contributions of Grauberger (2015) is that it does not need to use the players' complete payoff matrixes to compute the proposed approximate NE which would reduce tremendous time and is more suitable to solve a real-size game problem. Grauberger (2015) proposed the definition of a reduced game which is based on the strategies remaining available to the players after the elimination. An iterative search for best response is used to find a reduced game. The iterative search is terminated when the best response is selected twice time. When a termination takes place, there are two possible cases. If the chosen strategy is same as the one in the previous iteration, an NE is found since the strategies are mutually the best responses to each other. In the second case, if the same strategy is chosen in more than one iteration ago, the iteration search terminates to avoid sticking into an endless looping. Thus, the strategies which are selected twice are left and compose a reduced game. While the other strategies will be eliminated. It is worth noting that the stability of player's best response is significant. The reason is that the precondition of finding a pure NE or forming a reduced game is to guarantee the uniformity of the best response for each player (i.e. the best response must be a globally optimal solution) when facing the same competitor's strategy in different iterations. Hence, we need to use a global optimization algorithm to solve the best response.

Based on the definitions of the reduced game and ε-approximate NE, we define the ε-approximate NE in the reduced game as a new approximate NE for the original game. The approximate NE has three advantages. Firstly, the proposed approximate NE satisfies the decision-making process in the real market. When players cannot find NE strategies in the competition and trap into a looping, a feasible solution is to find acceptable strategies for both players (i.e. ε-approximate NE) in the reselected strategies. Secondly, when computing the approximate NE, the reduced game guarantees the computability of the approximate NE. Compared with the enormous numbers of strategy combinations, there is a relatively small number of strategies (even fewer than a dozen) left in the reduced game. Thirdly, the definition of ε-approximate NE makes sure that there exists a ε-approximate NE in the games with discrete strategies. So, a proposed approximate NE could be always found if required. When more than one equilibrium is found, the equilibrium with dominant payoff will be a priority, chosen as the final solution. If there is no payoff dominant equilibrium, the risk dominant equilibrium will be chosen as the final solution.

According to the notion of the proposed approximate NE, we develop an algorithm to find the solution. The algorithm is presented in Fig. 2.

5. Case study

We offer a case study about the competition between Dalian port (denoted as p_1) and Yingkou port (denoted as p_2) in the northeast of China to test the proposed game model and algorithm. Meanwhile, based on the results, we also investigate the impact of environmental concerns on the intermodal network design, profits variance, and CO2e emission mitigation.

The hinterland of Dalian port and Yingkou port is overlapping and covers three provinces (Liaoning, Jilin, and Heilongjiang) and the eastern region of Inner Mongolia. The 37 considered cities, which are listed in Table 2, include prefecture-level cities and the important transportation hub Meihekou in the study area. In Table 2, the first column presents the rank of a city with respect to container maritime transport demands which are obtained based on real-life data in 2015. The second and third columns exhibit the name of each city and corresponding identification number. The fourth column exhibits the percentage of container maritime transport demands. Based on the consideration of transport demands and traffic location of each city, 14 cities are selected as the candidate locations of dry ports, as shown in Fig. 3.

Table 3 shows the number of container berths and shipping lines. Assuming two ports have the same cost of port services and dry port services. The values of parameters about the cost of port and dry port service are measured based on a report of the construction of a terminal in Dalian port. The cost of road transport is calculated depending on distance multiplied by transport cost per TEU per kilometer, which is set as 6 CNY/(km·TEU). The cost of rail transport is obtained from the website of China Railway (source: http://www.12306.com). The transport time is calculated according

Step 1. Determining the starting player and strategy. Choose a starting player denoted by p_0. Another player is denoted by p^-. Randomly select a starting strategy $S_{p^-}^0$ of player p^- in iteration 0. The starting strategy satisfies (21), (23).

Step 2. Choosing the best response iteratively for the players. Execute the following loop until satisfying the break condition.

Loop

for j=1 to 2

$p \leftarrow p_j$;

If $j == 1$, $\tilde{k} \leftarrow k-1$; otherwise $\tilde{k} \leftarrow k$;

Given the competitor's strategy $S_{p^-}^{\tilde{k}}$, solve the ITNDPS model to find the optimal strategy S_p^k using **a branch and bound algorithm** described in **Section 4.2**. Calculate the payoff Π_p^k based on (16);

If $k > 1$, go to step 3;

If j==1, $k \leftarrow k+1$;

Step 3. Checking up the terminating condition of Nash equilibrium. Check whether the best strategy S_p^k payoff Π_p^k has occurred before.

For $l = 1$ to $k-1$

If $\Pi_p^k == \Pi_p^l \,\&\&\, S_p^k == S_p^l$

If $l == k-1$

The pure NE is found. If $j == 1$, $S_p^E \leftarrow S_p^l$ and $S_{p-}^E \leftarrow S_{p-}^{l-1}$; otherwise, $S_p^E \leftarrow S_p^l$ and $S_{p-}^E \leftarrow S_{p-}^l$. Break the loop in Step 2. The pure NE is S_p^E and S_{p-}^E

else

The pure NE doesn't exist. Let $l' \leftarrow l$ and $j' \leftarrow j$. Break the loop in Step 2. Go to Step 4 to find an approximate NE.

Step 4. Setting up a reduced game. For any player $p' \in P$, a strategy set $\mathbf{S}_{p'}^r$ can be generated from l th loop to $k-1$ th loop. Calculate the payoff, $\forall p \in P$, $S_p \in \mathbf{S}_p^r$ and $S_{p-} \in \mathbf{S}_{p-}^r$, based on equation (16). Generate the payoff matrix, go to step 5.

Step 5. Finding an approximate Nash equilibrium solution. For any given combination strategy (S_p, S_{p-}), $\forall S_p \in \mathbf{S}_p^r$ and $\forall S_{p-} \in \mathbf{S}_{p-}^r$, find dominate strategy S_p^* for any player $p \in P$ satisfying $\Pi_p(S_p^* \mid S_{p-}) \geq \Pi_p(S_p^{'} \mid S_{p-}), \forall S_p^{'} \in \mathbf{S}_p^r$. Calculate ε for each strategy combination (S_{p_1}, S_{p_2}), for p_1, $\varepsilon_1(S_{p_1}, S_{p_2}) = \Pi_{p_1}(S_{p_1}^* \mid S_{p_2}) - \Pi_{p_1}(S_{p_1} \mid S_{p_2})$ $\varepsilon_0 = \Pi^{p_0}(S_0^* \mid S_1) - \Pi^{p_0}(S_0 \mid S_1)$, for p_2, $\varepsilon_2(S_{p_1}, S_{p_2}) = \Pi_{p_2}(S_{p_2}^* \mid S_{p_1}) - \Pi_{p_2}(S_{p_2} \mid S_{p_1})$, $\varepsilon(S_{p_1}, S_{p_2}) = \max\{\varepsilon_1, \varepsilon_2\}$. Find the approximate NE, $(S_{p_1}^E, S_{p_2}^E) = \arg \min_{S_{p1} \in \mathbf{S}_{p1}^r, S_{p2} \in \mathbf{S}_{p2}^r} \varepsilon(S_{p_1}, S_{p_2})$.

Fig. 2. Algorithm for an approximate Nash equilibrium.

to the distance divided by velocity, which is set as 80 km/h (by road) and 30 km/h (by rail). The service quality v_p of Dalian and Yingkou port are 8 and 5. The degree of convenience of Dalian and Yingkou Port are set as 7 and 4 respectively. We adopt the following figure to convert CO2e emissions into monetary. The value of time is 39 CNY/h (= 5 USD/h) which is also used by Wang and Meng (2011). The used figure is 680 CNY/ton (= 100 USD/ton) which is also adopted by Qu et al. (2016). We consider the number of dry ports for each port is no more than 7, which means 49 combinations are taken into consideration in this case study. Note that the value of parameter to capture shippers' concerns in our case study should include 0, which means shippers are not going to consider environmental implication. Any other positive values can be selected according to different circumstances. In our case study, we take five integers from 0 to 5 to test our model. The parameter used to reflect port operator's concerns on environment takes the values {0, 0.2, 0.4, 0.6, 0.8, 1}. The related parameters are shown in Table 4.

The models and algorithm are implemented in C++ with

Table 2
Cities in the northeast of China date set.

Rank	ID (#)	City	Percent of Total Demand (%)	Rank	ID (#)	City	Percent of Total Demand (%)
1	1	Dalian	29.0	20	22	Panjin	0.8
2	3	Shenyang	19.7	21	16	Fushun	0.7
3	4	Changchun	7.7	22	21	Liaoyang	0.7
4	10	Harbin	6.6	23	25	Huludao	0.7
5	2	Yingkou	5.6	24	26	Liaoyuan	0.6
6	11	Daqing	4.4	25	35	Heihe	0.6
7	12	Mudanjiang	2.3	26	28	Baishan	0.6
8	5	Jilin	2.1	27	24	Chaoyang	0.6
9	15	Anshan	1.7	28	32	Shuangyashan	0.6
10	18	Dandong	1.4	29	8	Baicheng	0.6
11	14	Qiqihar	1.3	30	30	Jixi	0.5
12	17	Benxi	1.3	31	23	Tieling	0.5
13	7	Songyuan	1.3	32	20	Fuxing	0.3
14	36	Suihua	1.2	33	31	Hegang	0.3
15	29	Yanbian	1.2	34	37	Daxinganling	0.3
16	33	Jiamusi	1.2	35	13	Yichun	0.3
17	19	Jinzhou	1.2	36	34	Qitaihe	0.2
18	6	Siping	1.0	37	9	Meihekou	0.0
19	27	Tonghua	0.9				

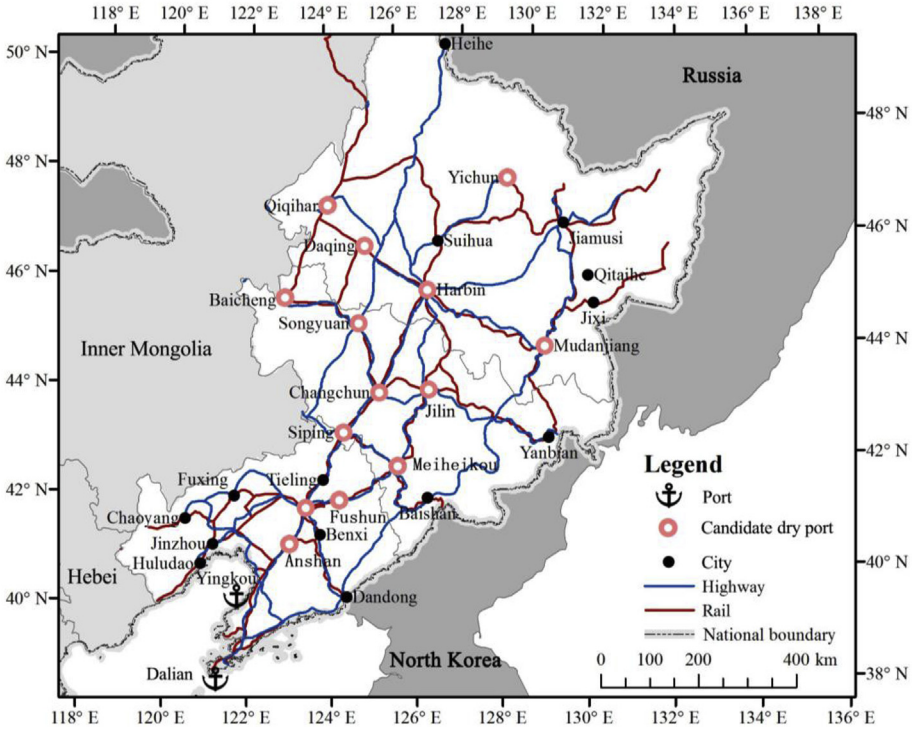


Fig. 3. Location map of data area.

Table 3
Information about Dalian port and Yingkou port.

Port	The number of berths	The number of shipping lanes
Dalian port	14	108
Yingkou port	8	70

Parallel Patterns Library. We use a PC with a 2.1 GHz E5-2620v2 processor and 16 GB of RAM, and operating system Win 7.

5.1. Port operators' equilibrium strategies

According to the value setting of the parameter, there are 6 values of the parameter κ and μ respectively, and 7 dry ports at most for each port competitor. Thus, we generate 1764 (product of the number of all variables) instances to include all scenes and test in the proposed model and algorithm. Each instance runs 10 times with different random starting strategy of the competitor. There are 127 instances in which the final results are approximate NE solutions. The number of these instances accounts for 7.2% of total instances. Table 5 shows the results in the instances in which the values of parameters used to reflect shippers' and port operator's concerns on the environment are fixed (i.e. $\kappa = 3, \mu = 2$). The results include the equilibrium strategies of port operators and the corresponding CO2e emissions releasing from freight transportation.

The selected locations shown in Table 5 have two characteristics which are same as the intuitive recognition. Firstly, the selected

locations, such as Harbin (10), Changchun (4) and Daqing (11), are mostly far from the ports more than 400 km are all satisfy the advantageous distance of intermodal transport. It helps the ports attracting more freights based on the cost advantage and environmental advantage of intermodal transport. Secondly, the locations with large transport demands are preferred by port operators, such as Changchun (4), Harbin (10), Daqing (11) and Jilin (5).

Furthermore, we investigate the effect of shippers' and port operators' environmental concerns on the dry port locations. In order to more comprehensively and clearly illustrate the effect, we make a statistic on the selection of dry port locations in the instances with the same stakeholders' environmental concerns and use the selection probability of each node to present the port operator' preference on each candidate node. When the values of κ and $\varepsilon$are fixed, there are 49 instances (7 alternative numbers of dry ports for each port operator), of which the set is denoted as $L_{\kappa,\mu}$. The selection probability of node h in $L_{\kappa,\mu}$ (denoted as$\xi^p_{h,\kappa,\mu}$) can be calculated by the following equation $\xi^p_{h,\kappa,\mu} = \left(\sum_{l\in L_{\kappa,\mu}} y^p_{h,l}\right)/49$, where $y^p_{h,l}$ is 1 when port operator p selected the node n as a dry port location in the instance l; 0, otherwise. Fig. 4 shows the selected probability of each candidate node when the value of κ is fixed (i.e.$\kappa = 2$). The blue pies represent the choice of Dalian port and the purple ones represent the choice of Yingkou port.

It is found that when μ is no more than 0.4, as shown in Fig. 4(a), (b), (c), (g), (h) and (i), the change of port operators' environmental concerns does not substantially influence the selection of dry port locations. Both ports prefer the candidate nodes in the north central of the region, especially Harbin (10), Daqing (11) and Changchun (4). Moreover, candidate nodes at greater distances have a higher possibility of being chosen. For example, Dalian port chooses Qiqihar (14), while Yingkou port chooses Mudanjiang (12).

Nevertheless, if μ is more than 0.6, the port operators will obviously change the selected locations. Especially, Dalian port will move the selected locations over a large range from the north to the south and avoid choosing the locations same as Yingkou port. As shown in Fig. 4(e), when$\mu = 0.8$, the selection probability for Dalian port of Harbin (10) and Shenyang (3) are respectively 12% and 43%, while for Yingkou port, corresponding selection probabilities are 88% and 12% (as shown in Fig. 4(k)). When $\mu = 1$ (i.e. the port operators only consider the environment when making a decision), For Dalian port, Harbin (10) is no longer a choice, and the probability reduces to 0%. While there is an increase in the probability of Shenyang (3). The probability of Shenyang (3) increase to 65% (Fig. 4(f)). For Yingkou port, the selection probability of Harbin (10) is 100%, while the probability of Shenyang (3) is 8% (Fig. 4(k)). Secondly, both ports prefer closer candidate nodes. For instance, Dalian port reduces the selection probability of Harbin (10), Daqing (11) and Changchun (4) which are all relatively far apart, and Yingkou port reduces the selection probability of Mudanjiang (12).

Fig. 5 shows how the shippers' environmental concerns affect selection of dry port locations. We focus the instances in which the value of the parameter used to reflect the environmental concerns of port operators is fixed (i.e. $\mu = 0.4$). With the increase of values of κ (i.e. shippers gradually pay more attention to environment), most nodes' selection probabilities have no evident changes between two ports. For example, the selection probabilities of Harbin (10) Changchun (4) and Daqing (11) for Dalian and Yingkou port are the same, and the values are respectively 100%, 71% and 86 when $\kappa = 0$, as shown in Fig. 5(a) and (g), then change to 98% and 86% and 71% for Dalian port, and 100% and 71% and 86% for Yingkou port when $\kappa = 5$ (Fig. 5(f) and (i)). On the other hand, with the increase of values of κ, both ports are more likely to choose eastern cities than western cities. The selection probability of Qiqihar (14) for Dalian port reduces from 57% ($\kappa = 0$) to 24% ($\kappa = 5$). From the perspective of Yingkou port, the increase of κ leads to the preference to the location closer to the port.

5.2. Port operator's profits

To explore how the environmental concerns of stakeholders influence these two port operators' profits, we compare the results with different combinations of levels of stakeholders' environmental concerns. Fig. 6 shows the profits of both ports for the problem with fixed numbers of dry ports for both ports (i.e. $n_{p_1} = 5$, $n_{p_2} = 5$). In Fig. 6 the value of the parameter used to reflect port operators' environmental concerns (μ) is shown on the horizontal axis and the level of shippers' environmental concerns (κ) is represented by different shades of a color. The blue points and purple points represent the profits of Dalian port and the profits of Yingkou port respectively. It is interesting to find that the rise of shippers' environmental concerns benefits the profits of Yingkou port which is closer to the market. As shown in Fig. 6, if the value of μ is

Table 4
Parameters used in application tests.

Parameters	Value	Unit
Parameter used to reflect shippers' concerns on environment (κ)	0, 1, …, 5	–
Parameter used to reflect port operator's concerns on environment (μ)	0, 0.2, …,1	–
The cost of port services of Dalian and Yingkou ports (c_p)	82	CNY/TEU
The cost of dry port services of Dalian and Yingkou ports (c^p_h)	50	CNY/TEU
Fixed cost of Dalian port (C_{p_1})	700	$\times 10^7$CNY
Fixed cost of Yingkou port (C_{p_2})	500	$\times 10^7$CNY
The charge of port services of Dalian and Yingkou port (τ_p)	800	CNY/TEU
The charge of dry port services of Dalian and Yingkou port (τ^p_h)	200	CNY/TEU
Scale parameter in transport model choice model (θ)	3.4	$\times 10^{-3}$
Value of time (ζ)	39	CNY per hour
Parameter used to identify the value related to the number of berths (η)	27	CNY per berth
Parameter used to identify the value related to the number of shipping lanes (ω)	13	CNY per route
Parameter used to identify the value of port service quality of Dalian and Yingkou port (γ)	76	CNY
Parameter used to identify the value of transport service quality of Dalian and Yingkou port (σ)	138	CNY
Number of dry ports for port p (n_{p_1},n_{p_2})	1, 2, …, 7	–
port service quality of Dalian port (v_{p_1})	8	–
port service quality of Yingkou port (v_{p_2})	5	–
The degree of convenience for road transport (z^{road})	7	–
The degree of convenience for railway transport (z^{inter})	4	–

Table 5
The equilibrium strategies of both ports.

n_{p_1}	n_{p_2}	Dalian Port Dry port location (#)	CO2e (10^3 t)	Yingkou Port Dry port location (#)	CO2e (10^3 t)
1	1	10	1647.60	10	1969.97
1	2	10	1502.26	4,10	1942.77
1	3	10	1368.65	4,10,11	1903.84
1	4	10	1191.54	4,10,11,12	1941.04
1	5	10	1141.82	4,5,10,11,12	1940.54
1	6	10	1110.53	4,5,7,10,11,12	1933.52
1	7	10	1093.40	4,5,6,7,10,11,12	1934.04
2	1	10,11	1758.21	10	1739.98
2	2	4,10	1544.51	4,10	1805.94
2	3	4,10	1422.00	4,10,11	1815.02
2	4	4,10	1247.17	4,10,11,12	1856.22
2	5	4,10	1196.48	4,5,10,11,12	1862.91
2	6	4,10	1162.61	4,5,7,10,11,12	1861.56
2	7	4,10	1144.97	4,5,6,7,10,11,12	1866.11
3	1	4,10,11	1803.14	10	1611.43
3	2	4,10,11	1636.29	10,11	1717.85
3	3	4,10,11	1512.80	4,10,11	1727.39
3	4	4,10,11	1337.97	4,10,11,12	1768.60
3	5	4,10,11	1296.48	4,7,10,11,12	1773.79
3	6	4,10,11	1245.92	4,5,7,10,11,12	1780.36
3	7	4,10,11	1207.91	4,5,7,10,11,12,14	1787.37
4	1	4,5,10,11	1804.87	10	1556.39
4	2	4,10,11,14	1685.44	10,11	1651.64
4	3	4,10,11,14	1562.28	4,10,11	1660.65
4	4	4,10,11,14	1387.45	4,10,11,12	1701.87
4	5	4,10,11,14	1348.46	4,10,11,12,14	1704.54
4	6	4,10,11,14	1297.90	4,5,10,11,12,14	1711.10
4	7	4,10,11,12	1240.16	4,5,7,10,11,12,14	1748.44
5	1	4,5,7,10,11	1833.63	10	1501.60
5	2	4,5,10,11,14	1687.15	10,11	1596.60
5	3	4,10,11,12,14	1527.79	4,10,11	1602.92
5	4	4,7,10,11,14	1415.78	4,10,11,12	1655.57
5	5	4,5,10,11,14	1368.80	4,10,11,12,14	1662.25
5	6	4,10,11,12,14	1330.15	4,5,10,11,12,14	1672.18
5	7	4,10,11,12,14	1282.45	4,5,7,10,11,12,14	1709.64
6	1	4,5,7,10,11,12	1787.74	10	1462.29
6	2	4,5,7,10,11,14	1713.33	10,11	1546.50
6	3	4,7,10,11,12,14	1556.06	4,10,11	1556.68
6	4	4,5,7,10,11,14	1436.07	4,10,11,12	1613.32
6	5	4,7,10,11,12,14	1397.26	4,7,10,11,12	1623.41
6	6	4,7,10,11,12,14	1347.90	4,7,10,11,12,14	1662.51
6	7	4,5,10,11,12,14	1302.96	4,5,7,10,11,12,14	1683.08
7	1	4,5,6,7,10,11,12	1803.57	10	1436.84
7	2	4,5,7,10,11,12,14	1667.48	10,11	1507.17
7	3	4,5,7,10,11,12,14	1570.38	4,10,11	1518.24
7	4	4,5,7,10,11,12,14	1465.29	4,10,11,12	1576.97
7	5	4,5,7,10,11,12,14	1425.68	4,7,10,11,12	1599.46
7	6	4,5,7,10,11,12,14	1378.24	4,5,7,10,11,12	1619.24
7	7	4,5,7,10,11,12,14	1330.47	4,5,7,10,11,12,14	1656.62

fixed, with the increase of the value of κ, the change of both port operators' profits present two opposite trends. Specifically, the profits of Dalian port will decline and the profits of Yingkou port will increase. Due to the geographical advantage of Yingkou port, if the shippers are more concerned with the environment when choosing transport routes, they will prefer Yingkou port due to the relatively close distance which helps release less CO_2e emissions. Thus, the increase of throughput of Yingkou port will improve the profits.

We also find that if the value of κ is fixed, the impact of μ on port operators' profits is related to the shippers' environmental concerns. As shown in Fig. 6, when the shippers' environmental concerns are weak (i.e.$\kappa = 0, 1$), the profits change slightly with the increase of μ, and obvious change can only be seen until $\mu > 0.8$. If environmental concerns are strong (i.e.$\kappa = 2, 3, 4$), the increase of μ causes significant changes to profits when $\mu > 0.6$. The profits of Yingkou port almost grows linearly, but the profits of Dalian port declines.

5.3. CO_2e emissions

In this subsection, we try to figure out the problem whether the construction of dry ports and the rise of stakeholders' environmental concerns will help reduce the CO_2e emissions releasing from the freight transport in the inland. Fig. 7 shows the total CO_2e emissions with different combinations of numbers of the dry ports when $\mu = 0.4$. The lighter color represents a higher value of κ which means shippers prefer the route with lower CO_2e emissions. It is found that the CO_2e emissions drastically decrease as the shippers pay more attention on the environment. As shown in Fig. 7, the stratification of CO_2e emissions obviously exists. CO_2e emissions in the instances in which $\kappa = 5$ averagely decrease by about 27.7% compared with the emissions in the instances in which $\kappa = 0$.

Fig. 7 also presents the effect of dry port construction on the reduce of CO_2e emissions. When the value of κ is fixed, the maximal amount of CO_2e emissions always appears in the instances in which both ports only plan to construct one dry port (i.e. $n_{p_1} = 1, n_{p_2} = 1$). When more dry ports are planned to be constructed, the CO_2e emissions will be reduced. While the effect of dry ports gradually declines as increasing the number of total dry ports. For example, when $\kappa = 1$, the CO_2e emissions in the instances where $n_{p_1} = 4, n_{p_2} = 4$ and $n_{p_1} = 7, n_{p_2} = 7$ decrease by 8.5% and 11.6% respectively compared with the emission amount in the instance where $n_{p_1} = 1, n_{p_2} = 1$. These results highlight the critical role of dry port construction in reducing CO_2e emissions.

Besides the effect of dry port construction, we are also interested how the rise of port operators' environmental concerns influence the total CO_2e emissions and the CO_2e emissions releasing from transporting freights to each port. In order to more clearly present the effect of port operators' environmental concerns, for each value of κ we generate instances in which $\mu \in \{0, 0.2, 0.4, \ldots, 1\}$ and $n_{p_1} = 5, n_{p_2} = 5$. Fig. 8 shows the results of CO_2e emissions in the aforementioned instances. The similar variation trend can be observed in the total CO_2e emissions and CO_2e emissions of Dalian port, while Yingkou port presents an opposite trend. The same phenomenon displayed is observed again that the rise of shippers' environmental concerns helps to reduce the total emissions. From the aspect of port operators' environmental concerns, with the increase of μ, as shown in Fig. 8(a)–(f), there is an apparent increase of Yingkou port's CO_2e emissions. For example, when $\kappa = 4$, comparing with Fig. 8(a)–(f), where the value of μ are 0 and 1 respectively, the CO_2e emissions of Yingkou port has increased over 11%. On the contrast, take $\kappa = 4$ as an example, the CO_2e emissions of Dalian port declines over 20% as μ changes from 0 to 1. The value of total CO_2e emissions hovers in $2600 \times 10^3 t$ to $2700 \times 10^3 t$ all the time, and does not show any significant change.

5.4. Summary

Intermodal transportation plays an important role in sustainable transportation, which has been well studied in many studies. Meanwhile, it could be intuitively understood that the rise of ports' and shippers' environmental concerns will promote the use of intermodal transport and indirectly influence intermodal network design, port operation, and GHG emissions which is important for the government to achieve carbon-reduction targets. However, in reality, intensified port competition, overlapping hinterland and shippers' route choice behavior make investigation on the effect of ports' and shippers' environmental concerns more difficult. Thus, a numerical analysis of the effect of environmental concerns is in urgent need. The results of the case study shed some lights on the problem. Some of the significant and interesting findings are summarized as follows:



Fig. 4. Selection probability for each candidate dry port location for Dalian port (blue) and Yingkou port (purple) when $\kappa = 1$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

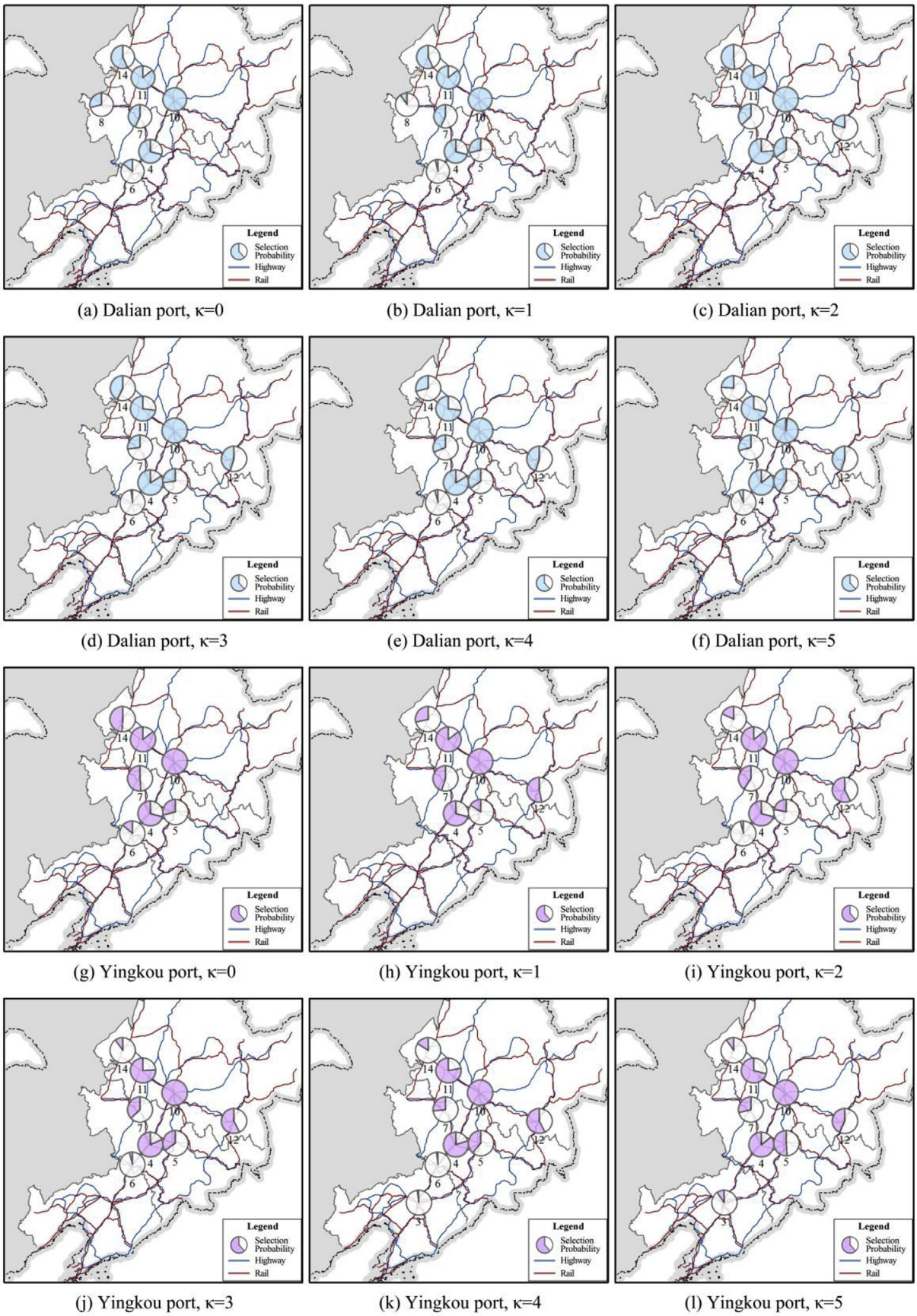


Fig. 5. Selection probability for each candidate dry port location for Dalian port (blue) and Yingkou port (purple) when $\mu = 0.2$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

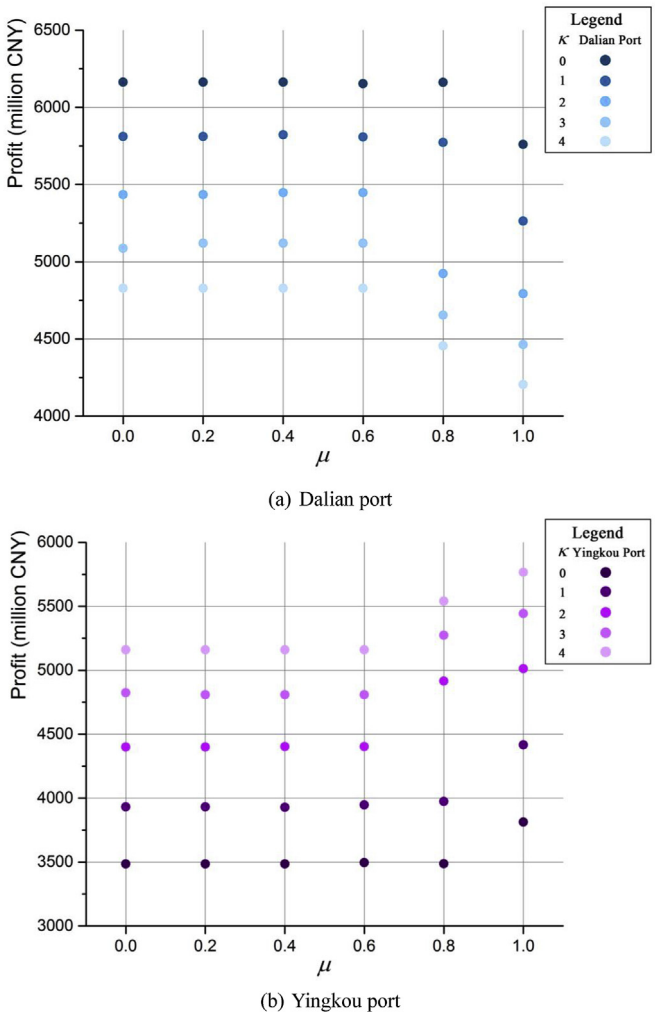


Fig. 6. Profits of port with different stakeholders' environment concerns.

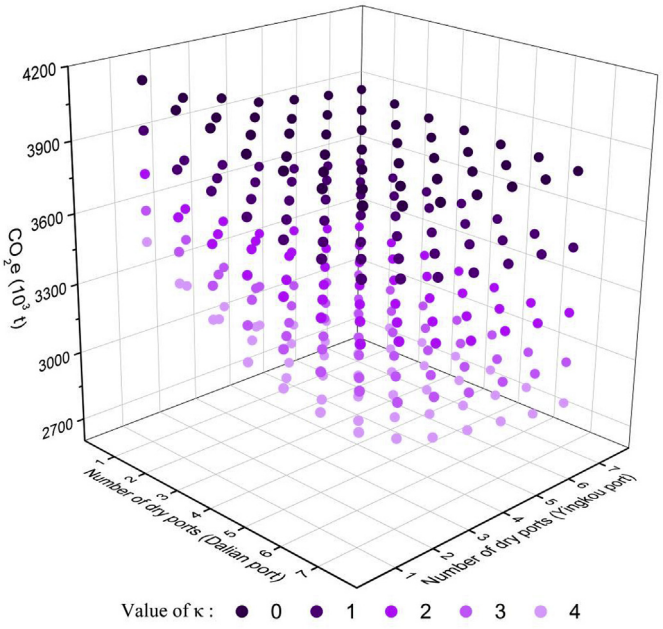


Fig. 7. CO_2e emissions with different numbers of dry ports ($\mu = 0.4$).

(1) The site selections of dry ports are closely related to their spatial locations and transport demands. Candidate nodes at greater distances with greater demands are more easily to be chosen as dry ports' locations. When the port operators' environmental concerns are weak, the increase of port operators' environmental concerns has limited influence on selections of dry ports. While the port operators' environmental concerns are strong, ports prefer to choose the cities closed to ports.

(2) There is a positive (negative) relationship between the shippers' environmental concerns and the profits of the port closer to (further from) market. The increase of port operators' and shippers' environmental concerns are beneficial to increasing Yingkou port's profits, because of regional advantages. In the contrast, the profits of Dalian port will decline with the increase of port operators' and shippers' environmental concerns.

(3) As the number of dry ports increases, the CO_2e emissions will be reduced. Meanwhile, the rise of shippers' environmental concerns will make the shippers choose the routes with lower emissions to reduce total CO_2e emissions, which will contribute to the reduction of Dalian port's throughput and increase the throughput of Yingkou port. Nevertheless, the rise of port operators' environmental concerns help reduce CO_2e emissions of Dalian port, while the trend of the change of CO_2e emissions of Yingkou port is opposite to that of Dalian port.

6. Conclusions

In this paper, we study the intermodal transport network with port competition, route choice behavior of shippers, and environmental concerns of stakeholders (i.e. port operators and shippers). To the best of authors' knowledge, this is the first work to address the intermodal transport network design with environmental concerns and port competition. In this process, competitive port operators plan to decide the strategy including the number of dry ports and the dry port locations. The problem is formulated as a game-theoretical model combined with a logit model used to describe the route choice behavior of shippers. To find a solution of the model, based on the definition of ε-approximate NE and reduced game, we propose a new approximate Nash equilibrium and the corresponding algorithm. This proposed algorithm does not only avoid the probability of nonexistence of pure Nash equilibrium solution, but also provides a method for reducing the calculation amount. Besides, a branch and bound algorithm is used to find the best response of a player.

A case study about the competition between Dalian port and Yingkou port in the northeast of China is adopted to test the proposed model and algorithm. Based on the case study, we examine how environmental concerns of stakeholders influence on the equilibrium strategies, profits, and CO_2e emissions. The results show that when stakeholders pay more attention to the environment, both port operators will more probably select cities closer to ports as dry port locations, especially for the Dalian port which is further away from the market. In addition, the rise of stakeholders' environmental concerns has also been identified. Yingkou port will always gain more profits, however, the profits of Dalian port decreases in most instances. With respect to CO_2e emissions, the results highlight that the rise of shippers' environmental concerns will contribute to the reduction of total CO_2e emissions. Nevertheless, the rise of port operators' environmental concerns helps reduce CO_2e emissions of Dalian port.

The main contributions of this study can be summarized as follows:

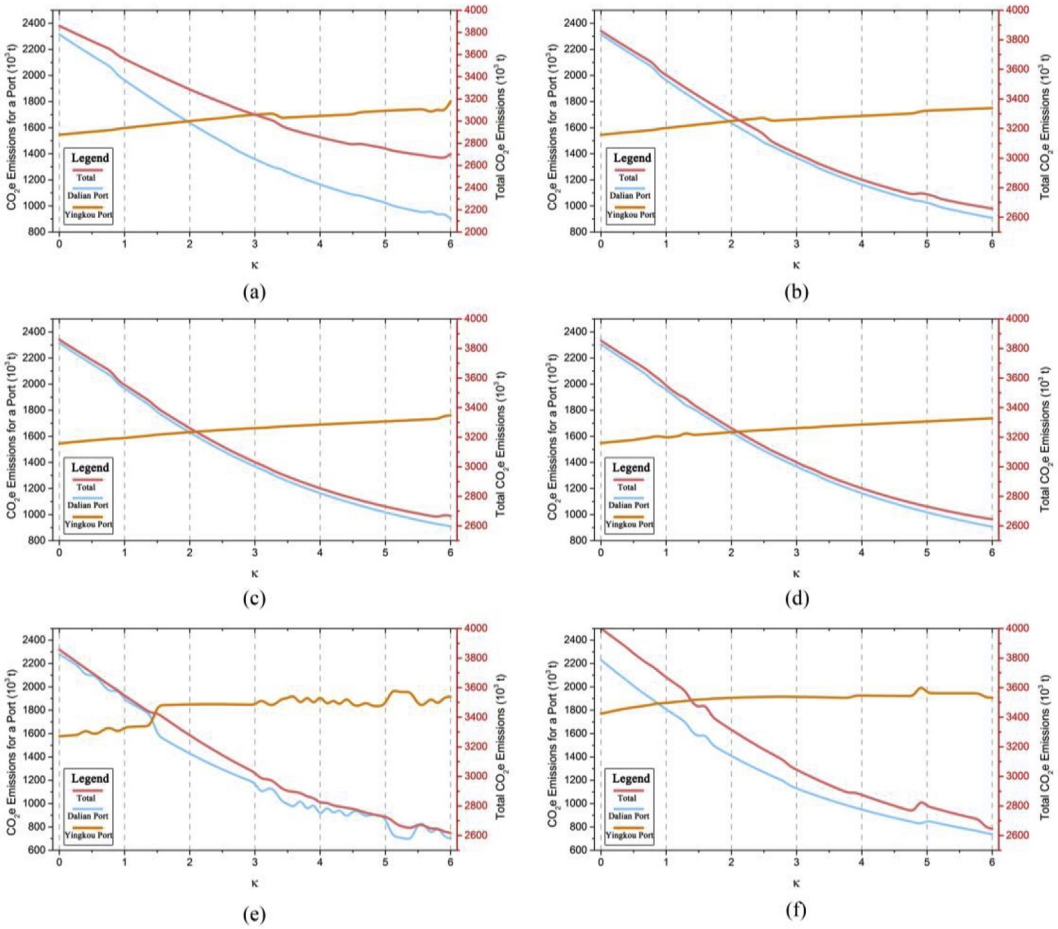


Fig. 8. CO_2 emissions with different environmental concerns for $n^p = 5, n^{p-} = 5$, (a) $\kappa = 0$, (b) $\kappa = 1$, (c) $\kappa = 2$, (d) $\kappa = 3$, (e) $\kappa = 4$, (f) $\kappa = 5$.

(1) It is the first time to study the low-carbon port intermodal network design from the perspective of games of ports with overlapping hinterland. A game model is introduced to find a Nash equilibrium solution on dry ports placement strategies to meet environment needs.
(2) We propose the ε-approximate NE in the reduced game, considering the probability of non-existence of pure Nash equilibrium solution. The proposed approximate NE satisfies the decision-making process in real market, and improves computational efficiency as well.

Further study can be conducted to explore governmental policies of reducing the environmental impacts: e.g., encouraging shippers to choose intermodal transport by providing subsidies or increasing the tax. It is also important to investigate situations in which port operators cooperate with each other and share dry ports to reduce the intensity of competition and the fixed costs. Moreover, it may be necessary to re-examine the problem based on a sequential game model, in which port operators make decisions in time order.

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References

Alumur, S., Kara, B.Y., 2008. Network hub location problems: the state of the art. Eur. J. Oper. Res. 190, 1–21.
Ambrosino, D., Sciomachen, A., 2016. A capacitated hub location problem in freight logistics multimodal networks. Optim. Lett. 10, 875–901.
Blum, A., Monsour, Y., 2007. Learning, Regret Minimization, and Equilibria, Algorithmic Game Theory. Cambridge University Press, New York, pp. 79–101.
Bosse, H., Byrka, J., Markakis, E., 2007. New algorithms for approximate Nash equilibria in bimatrix games. In: Deng, X., Graham, F.C. (Eds.), Lecture Notes in Computer Science, vol. 4858, pp. 17–29.
Chen, X., Deng, X., Teng, S., 2009. Settling the complexity of computing two-player Nash equilibria. J. ACM 56, 1–57.
Craig, A.J., Blanco, E.E., Sheffi, Y., 2013. Estimating the CO2 intensity of intermodal freight transportation. Transport. Res. Transport Environ. 22, 49–53.
Dekker, R., Bloemhof, J., Mallidis, I., 2012. Operations Research for green logistics - an overview of aspects, issues, contributions and challenges. Eur. J. Oper. Res. 219, 671–679.
Demir, E., Bektas, T., Laporte, G., 2014. A review of recent research on green road freight transportation. Eur. J. Oper. Res. 237, 775–793.
Demir, E., Bektaş, T., Laporte, G., 2011. A comparative analysis of several vehicle emission models for road freight transportation. Transport. Res. Transport Environ. 16, 347–357.
Demir, E., Burgholzer, W., Hrusovsky, M., Arikan, E., Jammernegg, W., Van Woensel, T., 2016. A green intermodal service network design problem with travel time uncertainty. Transp. Res. Part B Methodol. 93, 789–807.
Demir, E., Huang, Y., Scholts, S., Van Woensel, T., 2015. A selected review on the negative externalities of the freight transportation: modeling and pricing. Transport. Res. Part E Logist. Transport. Rev. 77, 95–114.
Eiselt, H.A., Marianov, V., 2009. A conditional p-hub location problem with attraction functions. Comput. Oper. Res. 36, 3128–3135.
Farahani, R.Z., Hekmatfar, M., Arabani, A.B., Nikbakhsh, E., 2013. Hub location problems: a review of models, classification, solution techniques, and applications. Comput. Ind. Eng. 64, 1096–1109.
Frémont, A., Franc, P., 2010. Hinterland transportation in Europe: combined transport versus road transport. J. Transport Geogr. 18, 548–556.

Geerlings, H., van Duin, R., 2011. A new method for assessing CO_2-emissions from container terminals: a promising approach applied in Rotterdam. J. Clean. Prod. 19, 657–666.
Grauberger, W., 2015. Network Revenue Management under Competition within Strategic Airline Alliances.
IFEU, 2016. Ecological Transport Information Tool for Worldwide Transports.
Ishfaq, R., Sox, C.R., 2010. Intermodal logistics: the interplay of financial, operational and service issues. Transport. Res. Part E Logist. Transport. Rev. 46, 926–949.
Ishfaq, R., Sox, C.R., 2011. Hub location-allocation in intermodal logistic networks. Eur. J. Oper. Res. 210, 213–230.
Kirschstein, T., Meisel, F., 2015. GHG-emission models for assessing the eco-friendliness of road and rail freight transports. Transp. Res. Part B Methodol. 73, 13–33.
Lam, J.S.L., Gu, Y., 2016. A market-oriented approach for intermodal network optimisation meeting cost, time and environmental requirements. Int. J. Prod. Econ. 171, 266–274.
Lättilä, L., Henttu, V., Hilmola, O., 2013. Hinterland operations of sea ports do matter: dry port usage effects on transportation costs and CO2 emissions. Transport. Res. Part E Logist. Transport. Rev. 55, 23–42.
Limbourg, S., Jourquin, B., 2009. Optimal rail-road container terminal locations on the European network. Transport. Res. Part E Logist. Transport. Rev. 45, 551–563.
Lin, C., Chiang, Y., Lin, S., 2014. Efficient model and heuristic for the intermodal terminal location problem. Comput. Oper. Res. 51, 41–51.
Lin, C., Lee, S., 2010. The competition game on hub network design. Transp. Res. Part B Methodol. 44, 618–629.
Lüer-Villagra, A., Marianov, V., 2013. A competitive hub location and pricing problem. Eur. J. Oper. Res. 231, 734–744.
Mahmutogullari, A.I., Kara, B.Y., 2016. Hub location under competition. Eur. J. Oper. Res. 250, 214–225.
Marianov, V., Serra, D., ReVelle, C., 1999. Location of hubs in a competitive environment. Eur. J. Oper. Res. 114, 363–371.
Meng, Q., Wang, X., 2011. Intermodal hub-and-spoke network design: incorporating multiple stakeholders and multi-type containers. Transp. Res. Part B Methodol. 45, 724–742.
Mostert, M., Limbourg, S., 2016. External costs as competitiveness factors for freight transport a state of the art. Transport Rev. 36, 692–712.
Notteboom, T.E., Rodrigue, J., 2005. Port regionalization: towards a new phase in port development. Marit. Pol. Manag. 32, 297–313.
Qu, Y., Bektaş, T., Bennell, J., 2016. Sustainability SI: multimode multicommodity network design model for intermodal freight transportation with transfer and emission costs. Network. Spatial Econ. 16, 303–329.
Resat, H.G., Turkay, M., 2015. Design and operation of intermodal transportation network in the Marmara region of Turkey. Transport. Res. Part E Logist. Transport. Rev. 83, 16–33.
Roso, V., Lumsden, K., 2010. A review of dry ports. Marit. Econ. Logist. 12, 196–213.
Sasaki, M., 2005. Hub network design model in a competitive environment with flow threshold. J. Oper. Res. Soc. Jpn. 48, 158–171.
Sasaki, M., Campbell, J.F., Krishnamoorthy, M., Ernst, A.T., 2014. A Stackelberg hub arc location model for a competitive environment. Comput. Oper. Res. 47, 27–41.
Sörensen, K., Vanovermeire, C., Busschaert, S., 2012. Efficient metaheuristics to solve the intermodal terminal location problem. Comput. Oper. Res. 39, 2079–2090.
SteadieSeifi, M., Dellaert, N.P., Nuijten, W., Van Woensel, T., Raoufi, R., 2014. Multimodal freight transportation planning: a literature review. Eur. J. Oper. Res. 233, 1–15.
U.S. Energy Information Administration, 2017. International Energy Outlook 2017.
Wagner, B., 2008. A note on "Location of hubs in a competitive environment. Eur. J. Oper. Res. 184, 57–62.
Wang, H., Meng, Q., Zhang, X., 2014. Game-theoretical models for competition analysis in a new emerging liner container shipping market. Transp. Res. Part B Methodol. 70, 201–227.
Wang, X., Meng, Q., 2011. The impact of landbridge on the market shares of Asian ports. Transport. Res. Part E Logist. Transport. Rev. 47, 190–203.
Wang, X., Meng, Q., 2017. Discrete intermodal freight transportation network design with route choice behavior of intermodal operators. Transp. Res. Part B Methodol. 95, 76–104.
Zhang, M., Wiegmans, B., Tavasszy, L., 2013. Optimization of multimodal networks including environmental costs: a model and findings for transport policy. Comput. Ind. 64, 136–145.